

Satrajit Ghosh

LinkedIn: [linkedin.com/in/satrajit-ghosh](https://www.linkedin.com/in/satrajit-ghosh)
GitHub: github.com/satrajitghosh183

Email: satrajitghosh183@gmail.com
Mobile: +1-640-250-7116
Portfolio: satrajitghosh.portfolio/satrajit-ghosh

Master's student in Electrical and Computer Engineering at Rutgers with research experience in software systems for AI, native ML tooling, document processing pipelines, retrieval and ranking infrastructure, and performance optimisation. Built large C++/Python systems for PDF parsing, equation OCR, structured LaTeX reconstruction, neural speech, search evaluation, and real-time simulation, with emphasis on debugging, benchmarking, reproducible experiments, and modular system design. Interested in software engineering spanning AI systems, testing and evaluation, program tooling, distributed workflows, and reliable high-performance software.

TECHNICAL SKILLS

Languages: C++, Python, Java, C#, JavaScript/TypeScript, SQL **Systems/Tools:** Git, Linux, CMake, SQLite, CUDA, OpenGL, ONNX Runtime, MuPDF

ML/AI Systems: LibTorch, PyTorch, OCR, retrieval/ranking, VLM pipelines, benchmarking, evaluation pipelines, experiment configuration

Research Areas: Software engineering for AI, AI for software engineering, systems reliability, testing and evaluation, program tooling

EDUCATION

Rutgers University New Brunswick, NJ
Master's in Electrical and Computer Engineering – GPA: 4.0/4.0 Sep 2024 – May 2026
◦ Relevant Coursework: Parallel and Distributed Computing, Computer Architecture, Reinforcement Learning, Multimodal Deep Learning

Princeton University Princeton, NJ
Graduate Exchange Program — Neural Rendering; Grade: A Spring 2025
◦ Research on NeRF-based avatar synthesis, view selection, and geometry-aware rendering workflows

Institute of Engineering and Management Kolkata, India
Bachelor of Technology in Computer Science and Business Systems – GPA: 9.28/10 Oct 2020 – Jul 2024
◦ Relevant Coursework: Object-Oriented Programming, DBMS, Automata Theory, Digital Electronics; 5th place Intel OneAPI, 1st place IEEE Elevate, 3rd place Google Solving for India

EXPERIENCE

Graduate Researcher Rutgers University
New Brunswick · NJ Jun 2025 – Present
◦ **Kuiper Project:** Architected a C++-centric system for academic PDF to structured LaTeX and speech, integrating MuPDF parsing, layout analysis, equation OCR, validation, and native inference workflows; co-authored two ISEC 2026 papers

◦ **iPapers.ai:** Built retrieval, ranking, and evaluation tooling for scientific search, including dataset preprocessing, parallel backend workflows, benchmark analysis, and large-scale result comparison

◦ **Sensing and Reasoning Lab:** Developed autonomous-driving scenario generation and benchmark execution workflows from real-world logs, supporting reliability-oriented simulation and evaluation

Graduate Grading Assistant Rutgers University ECE
New Brunswick · NJ Aug 2025 – Present
◦ Supported Computer Architecture, Convex Optimisation, and Intro to Machine Learning by debugging student code, writing solution keys, and providing detailed technical feedback on correctness, performance, and design

Tech & Product Intern Thirdbase Capital
New York · NY May 2025 – Sep 2025
◦ Built scoring and data-processing tools for private-capital analysis and developed full-stack internal prototypes with anomaly-detection components and model-backed workflows

AR/VR and ML Intern IEMA Research and Development
Kolkata · India May 2022 – Oct 2023
◦ Built AR/VR and mobile prototypes in Unity, Blender, and Android Studio, contributing interactive features, integration work, and deployment-focused debugging

PROJECTS

Kuiper-TTS — Academic PDF to Structured LaTeX and Speech
C++17, LibTorch, MuPDF, llama.cpp, ONNX Runtime, SQLite, React
◦ Built an end-to-end C++-centric research system for academic PDF ingestion, layout analysis, structured LaTeX reconstruction, equation OCR, and neural speech generation behind a unified native toolchain

◦ Implemented training and inference workflows with config-driven experimentation, evaluation metrics, HTTP services, and modular integration of external engines for document understanding and speech

LAGEngine — Master's Thesis Game Engine
C++17, OpenGL, CUDA, XPBD, SPH, PPO, Dear ImGui, Lua
◦ Built a unified real-time 3D engine with rigid-body dynamics, cloth and soft-body simulation, SPH fluids, deferred PBR rendering, ECS-based scene management, and a multithreaded work-stealing job system

◦ Integrated a passive PPO-based runtime tuner that reduced frame time by 23.79% and built educator-facing visual authoring tools for non-programmer users

Code & Conquer — Gamified Coding Platform
Node.js, WebGL2, WebSockets, PostgreSQL, Redis, OAuth
◦ Built core graphics and real-time architecture for a team-based coding platform, including WebGL2 rendering, matchmaking flows, WebSocket gameplay updates, and leaderboard-oriented backend design

Market Whisperer — News and Policy-Aware Trading Platform
Next.js, Express.js, MongoDB, Python, WebSockets
◦ Built a full-stack platform with REST APIs, live updates, quantitative analysis, sentiment-aware data processing, and model-generated explanations across backend and UI layers

Testing the McGurk Effect in Multimodal Machine Learning Models
PyTorch, Transformers, Audio-Visual Learning, Computer Vision
◦ Built congruent, mismatched, and adversarial audio-visual evaluation pipelines from the GRID corpus to test robustness of multimodal models under cross-modal sensory conflict

PUBLICATIONS

Satrajit Ghosh et al., “**Kuiper TTS: Making Legacy Scientific Papers Accessible by Converting Equations into Spoken Language**,” *Proc. ISEC 2026*, 2026. (2nd Prize, Best Paper)

K. G. August et al., “**Read the Math: Improving Mathematics Education Using Vision-Accessibility Techniques**,” *Proc. ISEC 2026*, 2026.

Satrajit Ghosh et al., “**VR Arena**,” *Springer Nature*, 2024.